FINAL PROJECT REPORT

Bachelor of Technology in Computer Science and Engineering

Class BTech CSE 25–29

PROJECT TITLE

Community Seed Bank

Preserving Local Seeds, Growing Stronger Communities

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Subject	GTM & Customer Operations

1. Executive Summary

The Community Seed Bank is a community-based agricultural resource model designed to help farmers access locally suitable seed varieties at planting time. The seed bank will collect, preserve, store and distribute small quantities of seeds to registered farmer members.

The proposed seed bank has a storage capacity of **2,000 seed packets** and expects approximately **600 member withdrawals per year**. Members pay **₹40 per packet**, while the average handling and packaging cost is **₹18 per packet**. This creates a basic contribution of **₹22 per packet** before other operating expenses.

The model focuses not only on selling seeds but also on creating a shared community resource. Farmers can deposit suitable local seeds, receive access to preserved varieties and participate in maintaining the seed bank. The system will use simple inventory records, quality checks, proper storage and seasonal demand planning.

The pilot will initially focus on a small group of farmer members, demonstrate the seed deposit-to-withdrawal process and then expand based on participation, seed quality and withdrawal demand.

2. Problem Statement and Opportunity

Problem Statement

Small and local farmers may face difficulty obtaining suitable seed varieties at the right time and in small quantities. Commercial seed sources may not always provide locally adapted varieties, while traditional local seeds can gradually disappear if they are not collected and preserved properly.

Farmers also need a simple and reliable system for:

- Accessing seeds during planting season
- Knowing which varieties are available
- Receiving seeds in small quantities
- Maintaining good-quality seed stock
- Preserving locally suitable varieties
- Keeping records of seed deposits and withdrawals

Opportunity Statement

A Community Seed Bank can convert locally available seeds into a shared community resource by collecting suitable varieties, storing them safely and making them available to registered members when required.

The opportunity is to build a **low-cost, community-supported seed access system** that combines seed preservation, seasonal inventory planning and member participation.

3. Target Customer Segment

Primary Customers

The primary customers are:

- Small and marginal farmers
- Local farmers growing seasonal crops
- Farmers interested in locally suitable varieties
- Community farming groups
- Farmer producer groups and cooperatives

Secondary Stakeholders

Other stakeholders can include:

- Agricultural volunteers
- Local agricultural organizations
- Seed suppliers
- Community leaders
- NGOs working with farmers
- Agricultural colleges and extension workers

Customer Profile

Example Farmer Persona:

Name: Ravi
Occupation: Small farmer
Requirement: Needs small quantities of reliable seed before planting season
Pain Points:

- Seeds may not be available locally at the required time
- Commercial packs may be larger than required

- Difficulty maintaining old/local varieties
- Uncertainty about seed quality
- Limited storage facilities

Needs: Affordable, locally suitable, accessible and properly stored seeds.

4. Value Proposition Canvas

Customer Jobs

Farmers need to:

1. Obtain seeds before planting.
2. Select suitable varieties.
3. Access small quantities.
4. Preserve useful local varieties.
5. Reduce wastage.
6. Plan seasonal cultivation.

Customer Pains

- Seed shortage during planting season
- Limited access to local varieties
- High cost of buying large seed packs
- Poor storage conditions
- Lack of information about available varieties
- Risk of receiving poor-quality seeds

Customer Gains

- Affordable seed access
- Locally suitable varieties
- Small quantity availability
- Better preservation of traditional seeds
- Convenient withdrawal process
- Community-level seed security

Products and Services

The Community Seed Bank provides:

- Seed collection
- Seed quality checking
- Cleaning and packaging
- Storage
- Inventory management
- Member registration
- Seasonal seed withdrawal
- Basic seed information

Pain Relievers

The model reduces problems through:

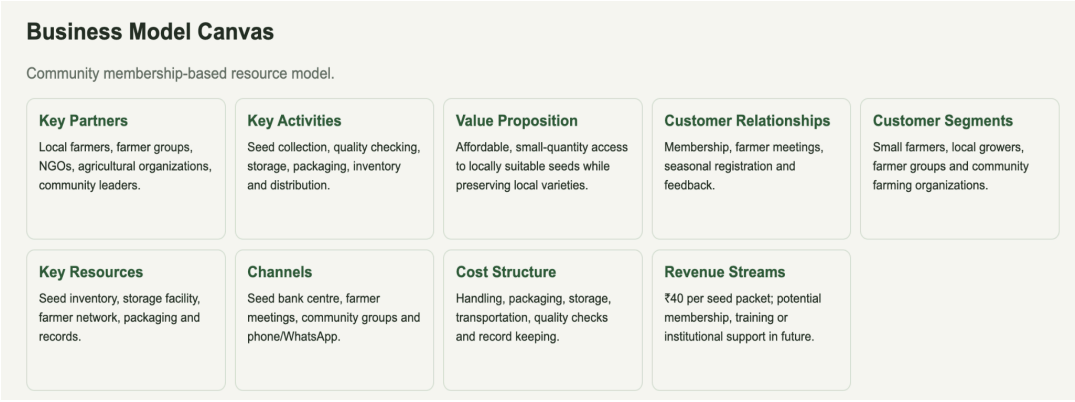
- Pre-season inventory planning
- Quality checks
- Proper labelling
- Controlled storage
- Member reservation
- Digital/basic inventory records

Gain Creators

The model creates value by:

- Preserving local varieties
- Making seeds available when needed
- Encouraging farmer participation
- Creating a shared agricultural resource
- Providing seeds in manageable quantities

5. Business Model Canvas



6. Seed Deposit and Withdrawal Journey

The seed bank should provide a simple end-to-end process.

Seed Collection → Quality Check → Cleaning → Packaging → Labelling → Storage → Inventory Entry → Member Request → Verification → Withdrawal → Record Update

Step 1: Seed Collection

Farmers deposit locally suitable seeds with the seed bank.

Information recorded:

- Farmer name
- Crop
- Variety
- Quantity
- Collection date
- Source/location

Step 2: Quality Check

Seeds are inspected for:

- Physical damage
- Moisture
- Pest/insect damage
- Visible contamination
- Basic germination quality

Poor-quality seeds are separated.

Step 3: Cleaning and Packaging

Accepted seeds are cleaned and packed into standard packets.

Each packet receives a label containing:

- Crop name
- Variety
- Batch number
- Storage date
- Basic usage information

Step 4: Storage

Packets are placed in suitable storage conditions and organized according to crop and variety.

Step 5: Inventory Entry

The stock record is updated after every deposit and withdrawal.

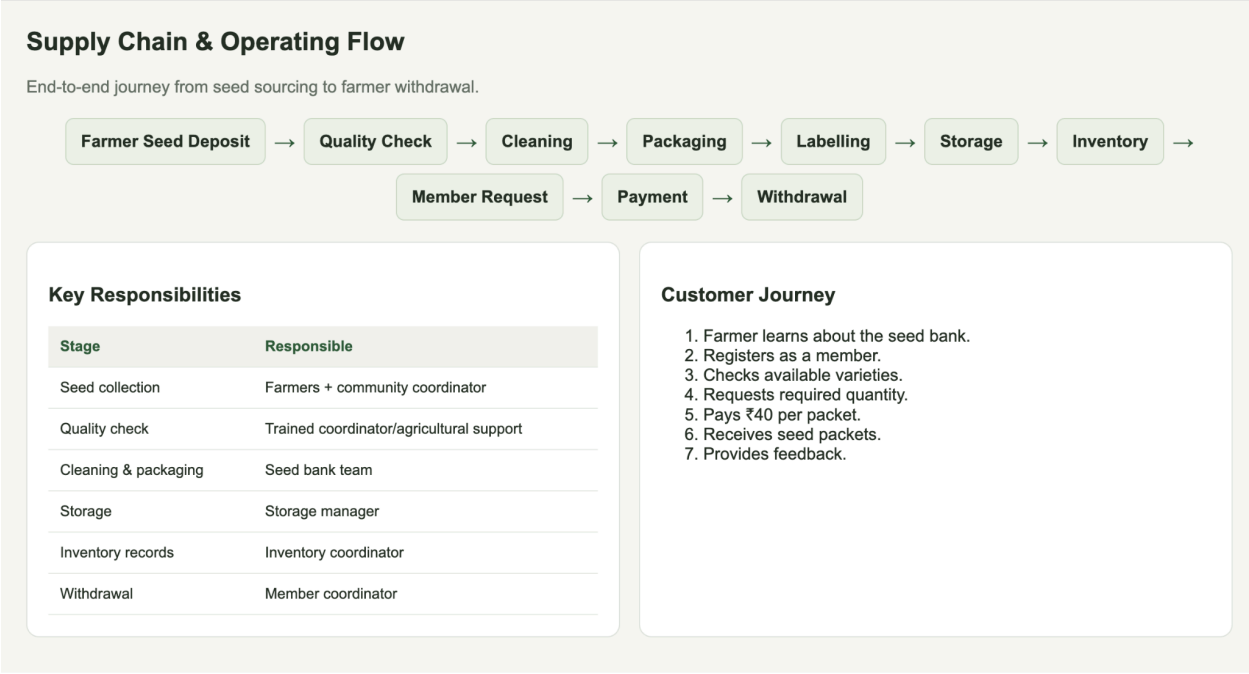
Step 6: Member Withdrawal

A registered member requests the required seed packet.

The staff/member coordinator:

1. Checks membership.
2. Checks availability.
3. Records withdrawal.
4. Collects ₹40 per packet.
5. Provides the seed packet.
6. Updates inventory.

7. Supply Chain / Operating Model



.8. Customer Journey and Service Touchpoints

Awareness

Farmers learn about the seed bank through farmer meetings, WhatsApp groups and local community leaders.

Registration

Farmers register as members and provide basic contact and farming information.

Seed Discovery

Members receive information about available crops and varieties.

Request

The farmer requests the required number of packets before planting.

Verification

The seed bank checks member status and inventory availability.

Payment

The farmer pays **₹40 per packet**.

Withdrawal

The seed packet is handed over and the transaction is recorded.

Feedback

The farmer provides feedback about availability, quality and germination performance.

9. Revenue and Cost Model

The case provides:

- Storage capacity = **2,000 packets**

- Annual withdrawals = **600 packets**
- Price per packet = **₹40**
- Handling and packaging cost = **₹18 per packet**

Annual Revenue

600 × ₹40 = ₹24,000

Therefore, expected annual packet revenue is:

₹24,000

Annual Handling and Packaging Cost

600 × ₹18 = ₹10,800

Contribution Before Other Costs

₹24,000 − ₹10,800 = ₹13,200

Therefore, the seed bank generates a basic contribution of:

₹13,200 per year

or

₹22 per withdrawn packet

This ₹22 is available to contribute toward other expenses such as storage, transportation, quality testing, record keeping and community activities.

> **Important:** ₹13,200 should not be treated as final profit because the case only provides handling and packaging cost. Other operating expenses still need to be considered.

Basic Performance Measures

The seed bank can track:

Annual withdrawal rate

600 ÷ 2,000 × 100 = **30%**

So, approximately **30% of the packet capacity is represented by annual withdrawals**, assuming the 2,000-packet figure is used as the comparison base.

Other KPIs:

- Number of active members
- Number of seed varieties preserved
- Number of packets withdrawn
- Seed stock availability
- Rejection rate during quality checking
- Repeat member participation
- Customer satisfaction
- Seed wastage percentage

10. Two Ways to Build Farmer Participation

1. Seed Contribution + Member Access

Farmers can contribute suitable local seeds to the seed bank.

In return, they become part of the community resource system and receive access to available seed varieties during planting seasons.

This encourages farmers to view the seed bank as a shared resource rather than simply a shop.

2. Farmer Demonstration and Community Events

The seed bank can conduct small demonstration sessions before planting season.

Activities can include:

- Seed selection demonstrations
- Storage awareness sessions
- Germination demonstrations
- Farmer meetings
- Local variety showcases

Farmers who see the process working may be more willing to participate and contribute seeds.

11. SWOT Analysis

Strengths

- Community-based model
- Locally suitable seed varieties
- Low-cost withdrawal model
- Helps preserve local seed varieties
- Creates shared community resources

Weaknesses

- Limited storage capacity
- Dependence on farmer participation
- Manual records may create errors
- Quality testing capability may be limited
- Seasonal demand can create uneven utilization

Opportunities

- Partnerships with agricultural organizations
- Expansion into more crop varieties
- Digital inventory system
- Farmer training programs
- Collaboration with farmer producer organizations

Threats

- Pest or moisture damage
- Poor seed quality
- Low farmer participation
- Seasonal demand fluctuations
- Extreme weather affecting seed availability

12. Key Risks and Mitigation

Risk	Impact	Mitigation
Moisture damage	Seed quality may decline	Use dry, controlled storage
Pest infestation	Loss of inventory	Regular inspection and pest-control procedures
Poor seed quality	Low farmer satisfaction	Quality inspection before storage
Low participation	Insufficient community resource	Awareness meetings and demonstrations
Stock shortage	Farmers cannot obtain desired variety	Seasonal demand forecasting
Record errors	Incorrect inventory	Maintain standardized digital/physical records
Seed variety loss	Local varieties disappear	Maintain reserve quantities and multiple contributors

13. Recommended Business Model and Go-to-Market Approach

The recommended model is a **community membership-based resource model**.

The seed bank should initially operate at a small scale instead of immediately trying to fill its entire 2,000-packet capacity.

Phase 1 – Community Formation

Identify farmers and understand:

- Crops grown
- Planting seasons
- Seed requirements
- Local varieties
- Current seed-access problems

Phase 2 – Seed Collection

Collect suitable seeds from participating farmers and perform basic quality checks.

Phase 3 – Pilot Distribution

Allow registered members to withdraw seeds at **₹40 per packet**.

Phase 4 – Feedback

Collect information regarding:

- Seed quality
- Availability
- Farmer satisfaction
- Germination experience
- Preferred varieties

Phase 5 – Expansion

Use the pilot results to increase farmer participation and gradually expand the number of varieties.

14. 90-Day Pilot-to-Scale Plan

Days 1–30: Setup

Activities:

- Identify 20–30 potential farmer members
- Conduct farmer interviews
- Identify important local varieties

- Arrange storage space
- Create inventory register
- Create membership form
- Collect initial seed deposits

Success Criteria:

- At least 20 registered members
- Initial seed inventory created
- Basic storage system operational
- Inventory records completed

Days 31–60: Pilot Operation

Activities:

- Complete quality checks
- Package and label seeds
- Start member withdrawals
- Record every transaction
- Conduct one farmer demonstration
- Collect feedback

Success Criteria:

- At least 50–100 packets properly recorded
- At least 20 member interactions
- 100% of withdrawals recorded
- No major inventory mismatch

Days 61–90: Evaluation and Scale

Activities:

- Analyze withdrawals
- Identify high-demand varieties
- Review farmer feedback
- Improve storage procedures
- Recruit additional members
- Plan next seasonal inventory

Success Criteria:

- 30+ active members
- Increased repeat participation
- Accurate inventory records
- Identified top-demand varieties
- Clear plan for expanding toward the 2,000-packet capacity

17. Conclusion

The Community Seed Bank provides a practical community resource model for preserving locally suitable seed varieties while improving seasonal seed access for farmers. The proposed model combines farmer participation, quality checking, storage, inventory management and controlled withdrawals.

With a capacity of 2,000 packets and an expected 600 annual withdrawals, the model can generate **₹24,000 in annual packet revenue** and **₹13,200 contribution before other operating expenses**, based on the figures provided in the case.

The 90-day pilot should focus on proving three things: **farmers are willing to participate, seeds can be stored and managed reliably, and members will use the service during planting periods.** Once these are demonstrated, the seed bank can gradually expand its farmer network, seed varieties and operating capacity.